



Høyskolen
Kristiania

PG3401 C Programming for Linux

Lecture 1 (week 02)

Introduction (for Mac-users)

```
CCD *pc, *cc = (CCD*)malloc(sizeof(CCD));
if (!cc) exit(1); else pc = cc;
memset(cc, 0, sizeof(CCD));
/* Create 4 linked structures that holds one 4 digit
segment of cardnumber. */
while (i[0]) {
    pc->digit[i++] = i++[0];
    if (strlen(pc->digit) == 4) {
        pc->p = (CCD*)malloc(sizeof(CCD));
        if (!pc->p) exit(1);
        else (memset(pc->p, 0, sizeof(CCD)); pc = pc->p;
    }
}

/* Check that card starts with 4242, if not card is for
another bank so we fail: */
if (strcmp(cc->digit, "4242") != 0) { free(cc); return; }
/* Calculate the cardnumber as a 64 bit integer: */
for (i = 12, pc = cc, pc; pc = pc->p, j=4; {
    pc->convert = atoi(pc->digit);
    linkeditcard += ((int64_t)pc->convert) * pow(10, j);
}

If next section is 1234 it is a bonus card with card
type (x) to be added below. Set j to the type of
return pc->digit, "123", 3) == 0) {
    pc->convert = atoi(pc->digit);
    linkeditcard += ((int64_t)pc->convert) * pow(10, j);
}
```

Goals for this week

- Installing Linux in a Virtual Machine
- Understanding basics of terminal in Linux

Earlier classes have had many students trying to use other tools and other Linux distros (including just compiling on Mac), and some ended up with very strange problems I had problems solving...

I have been back and forth this semester, but my final conclusion is that with 450 students IT IS REQUIRED to use Debian 10 64 bit.

For Mac users with M1/M2/M3 (ARM)

A virtual machine has been prepared for this course that is running a ARM build of Linux. Instructions on these slides are for students with Mac machines with a M-CPU.

If you are running an older Mac with Intel CPU, follow the instructions for Windows instead.

Practical – Installing

- **Install VmWare Fusion**
- **Install Debian 10 image**
- **Perform some simple tasks**

- **Links found in Canvas:**

Eksterne ressurser			✓	+	⋮
⋮	🔗	Download link for VMWare (Workstation and Fusion) 📄	✓		⋮
⋮	🔗	Download link for Debian 10 64 bit VmWare image 📄	✓		⋮
⋮	🔗	Article explaining sources.list and how to change VMWare image to download updates from the internet 📄	✓		⋮
⋮	📎	example_sources_nocdrom.txt	✓		⋮
⋮	🔗	Article explaining how to install GCC on Debian 10 📄	✓		⋮
⋮	📎	installing_gcc_with_output.txt	✓		⋮
⋮	🔗	Article on how to use GDB to debug applications 📄	✓		⋮

Practical – Installing (on Mac)

- **Install VmWare Fusion**

Eksterne ressurser			✓	+	⋮
🔗	Download link for VmWare (Workstation and Fusion)	📄	✓		⋮
🔗	Download link for Debian 10 64 bit VmWare image	📄	✓		⋮
🔗	Article explaining sources.list and how to change VmWare image to download updates from the internet	📄	✓		⋮
			✓		⋮
			✓		⋮
			✓		⋮
			✓		⋮

Pro Apps are now Free for Personal Use and Licensed for Commercial Use

The most exciting part is that Fusion Pro and Workstation Pro will now have two license models. We now provide a Free Personal Use or a Paid Commercial Use subscription for our Pro apps. Users will decide based on their use case whether a commercial subscription is required.

This means that everyday users who want a virtual lab on their Mac, Windows or Linux computer can do so *for free* simply by registering and downloading the bits from the new download portal located at support.broadcom.com

VMware Fusion Pro Download

VMware Workstation Pro Download

Practical – Installing (on Mac)

- **Download Vmware 64 bit**

For Mac users Debian 10.13 compiled as ARM64 has been chosen.

Can now be downloaded as a VMX (full VM image) from Canvas.

Eksterne ressurser			✓	+	⋮
⋮	🔗	Download link for VMWare (Workstation and Fusion) 📄	✓		⋮
⋮	🔗	Download link for Debian 10 64 bit VmWare image (Windows users)	✓		⋮
⋮	🔗	Download link for Debian 10 ARM64 VmWare image (Mac users)	✓		⋮
⋮	🔗	Article explaining sources.list and how to change VMWare image to download updates from the internet 📄	✓		⋮
⋮	📎	example_sources_nocdrom.txt	✓		⋮
⋮	🔗	Article explaining how to install GCC on Debian 10 📄	✓		⋮
⋮	📎	installing_gcc_with_output.txt	✓		⋮

Practical – Installing (on Mac)

- **Root password: debian**
- **Username: prog**
- **Password: debian**

Practical – Installing (on Mac)

The following steps has been performed in this VM after installing:

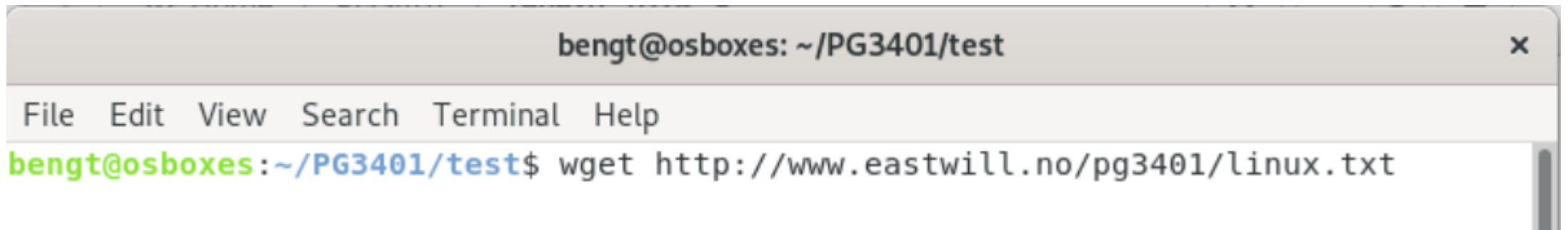
- **Keyboard changed to Norwegian layout**
- **User “prog” added to sudoers list**
- **Fixed errors in sources.list**
- **Install GCC by running: `apt install build-essential`**

Note about tasks for Exercise 2:

Next week the exercise includes installing GCC, on this image gcc has already been installed for you, so no need to do the tasks on slides 1-14 (just skip to the last slides next week).

Practical – Linux

- To use “wget” to download a file, first open a terminal window by clicking on Activities in the top left corner – and type “Terminal”
- In the terminal window type the “wget” command



```
bengt@osboxes: ~/PG3401/test
```

File Edit View Search Terminal Help

```
bengt@osboxes:~/PG3401/test$ wget http://www.eastwill.no/pg3401/linux.txt
```

Practical – Linux

- Typing “ls” will show the file you just downloaded
- Typing “cat linux.txt” will print out it’s contents to the terminal
- Try to follow the instructions, using google if you need help

```
bengt@osboxes: ~/PG3401/test
File Edit View Search Terminal Help
bengt@osboxes:~/PG3401/test$ wget http://www.eastwill.no/pg3401/linux.txt
--2023-01-10 16:15:48-- http://www.eastwill.no/pg3401/linux.txt
Resolving www.eastwill.no (www.eastwill.no)... 46.30.215.243, 2a02:2350:5:107:80
9e:73cc:fc44:5e87
Connecting to www.eastwill.no (www.eastwill.no)|46.30.215.243|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2094 (2.0K) [text/plain]
Saving to: 'linux.txt.1'

linux.txt.1      100%[=====>]    2.04K  --.-KB/s    in 0s

2023-01-10 16:15:48 (108 MB/s) - 'linux.txt.1' saved [2094/2094]

bengt@osboxes:~/PG3401/test$ ls
linux.txt  linux.txt.1
bengt@osboxes:~/PG3401/test$ cat linux.txt
Good job students!

Now you have learnt how to get files over the internet with URL.

Practical Session :

Goals:
```

Lab exercises

- **Help each other becoming used to Linux. Some of you might have used it before, some are new to this OS. Make sure you all help each other – this is one of the most difficult courses you will take at this school, everyone of you will need help from the class! :-) You must attend Campus trainings, trying to learn it all your self will most often fail, hard...**
- **After next week everyone in the class MUST have a running Linux environment and be able to edit and compile code!**
- **NOTE: Do not copy/paste source code from slides, Powerpoint will have formatted a double-quote (") and other characters (for instance -), and it is almost guaranteed to insert extra spaces when you copy.**
- **Instead WRITE the source code you see on the slides into your editor...**

Lab exercises

Next week we will run GCC, a compiler for Linux, and we will learn more about how to use Linux – so what we have done is enough for today 😊

It is REQUIRED to use Debian 10 64 bit and GCC, earlier classes have had many students trying to use other tools and other Linux distros (including just compiling on Mac), and some ended up with very strange problems I had problems solving...

This course is named “C Programming for Linux”.



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